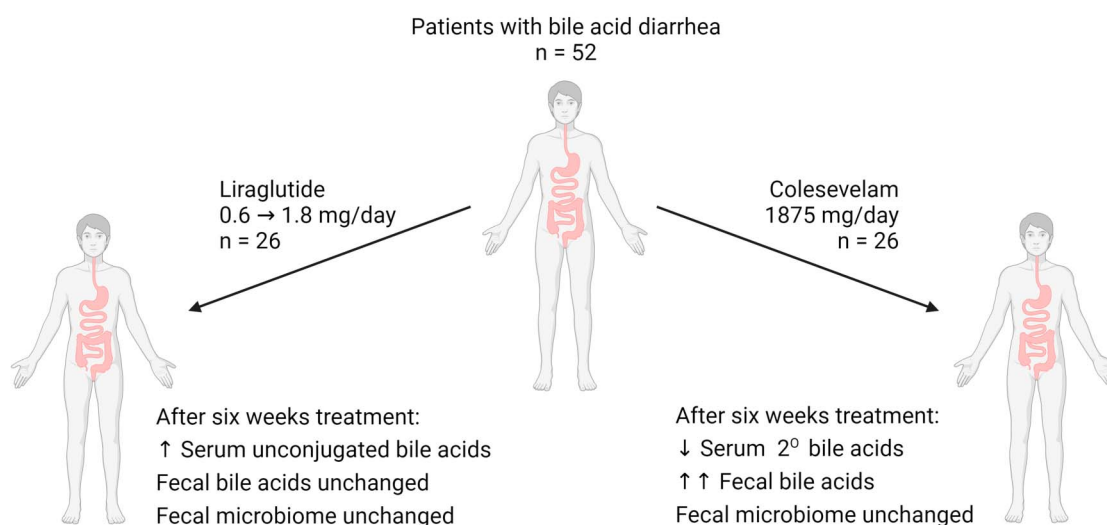


Liraglutide and Colesevelam Change Serum and Fecal Bile Acid Levels in a Randomized Trial With Patients With Bile Acid Diarrhea

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INTRODUCTION: Both liraglutide and colesevelam improve bile acid diarrhea symptoms. Colesevelam binds excess amounts of diarrhea-causing bile acids in the colon, whereas the mode of action for liraglutide remains elusive. In this article, we examined the impact of colesevelam and liraglutide treatment on the concentrations of bile acids in serum and feces and the fecal microbiota composition to better understand the 2 drugs' modes of action.

Liraglutide and colesevelam changes serum and fecal bile acid levels in a randomized trial with patients with bile acid diarrhea



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- METHODS:** Bile acid species were analyzed in serum and fecal samples from a randomized, double-blind, double-dummy trial at baseline and after 3 and 6 weeks of orally administered colestevlam (1,875 mg twice daily, $n = 26$) or subcutaneously administered liraglutide (up-titrated by weekly increments of 0.6 mg from 0.6 to 1.8 mg daily, $n = 26$) in patients with 75 selenium-homotaurocholic acid test-verified, idiopathic, or postcholecystectomy bile acid diarrhea. Fecal microbiota composition was analyzed by 16S rRNA gene amplicon sequencing at the same time points.
- RESULTS:** Colestevlam increased the fecal concentrations of all bile acid species, whereas it decreased serum concentrations of secondary bile acids. Liraglutide induced a small increase in serum unconjugated bile acid concentrations without affecting fecal bile acid concentrations. No changes in fecal microbiota composition were observed with either treatment.
- DISCUSSION:** Colestevlam and liraglutide exhibit distinct effects on serum and fecal bile acid concentrations with colestevlam reducing serum concentrations of secondary bile acids and promoting fecal bile acid excretion, whereas liraglutide enhances serum concentrations of unconjugated bile acids, potentially through deceleration of small intestinal transit time allowing more time for passive absorption of bile acids.

KEYWORDS: bile acid diarrhea; bile acid malabsorption; bile acid; liraglutide; colestevlam; glucagon-like peptide-1 receptor agonist; bile acid sequestrant; microbiome; enterohepatic circulation

SUPPLEMENTARY MATERIAL accompanies this paper at <http://links.lww.com/CTG/B205>

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INTRODUCTION

Bile acid diarrhea (BAD) is a socially debilitating disease with primary symptoms of diarrhea, abdominal pain, urgency, and fecal incontinence. Around 1% of the population is estimated to suffer from the disease (1,2), which is, nevertheless, relatively unknown. The symptoms are caused by an excessive spillover of bile acids from the ileum to the colon (because of the overproduction of bile acids and/or reduced absorption of bile acids in the small intestine). This irritates the colonic mucosa and induces peristalsis and osmotic water retention in the lumen (3). BAD can arise with no underlying cause, termed idiopathic BAD, or because of diseases, surgery, or other conditions, termed secondary BAD.

The intestinal microbial population, predominantly located in the colon, deconjugates the conjugated bile acids and converts the primary bile acids into secondary bile acids by dehydroxylation to protect themselves from the detergent effect of the bile acids (4,5). The bactericidal effect of the different bile acids varies dramatically with the secondary bile acid deoxycholic acid as the most potent (6). Thus, the composition of the bile acid pool can modulate the microbiome composition and vice versa (7–10).

BAD is most commonly treated with bile acid sequestrants tightly binding the bile acids in the intestinal lumen, which in turn reduces the colonic irritation and, thus, alleviates the symptoms (11). The sequestration prevents bile acid reuptake leading to excessive fecal excretion of bile acids (11–13) and a diminished bile acid serum pool triggering compensatory bile acid synthesis (12–14). As such, the treatment diminishes the symptoms but does not correct the underlying pathophysiology of BAD.

Recent studies show that treatment of patients suffering from BAD with the bile acid sequestrant colestevlam does not change the species richness of the gut microbiome but changes the abundance of certain species (15,16). The consequences of these changes remain unclear, but the bacterial composition of the microbiome may nevertheless predict treatment response to colestevlam (15).

Recently, we published a randomized, double-blind, double-dummy, noninferiority study showing that 6 weeks of treatment with the glucagon-like peptide 1 receptor agonist liraglutide reduced the number of daily bowel movements $\geq 25\%$ in more patients with primary BAD than colestevlam did (12,17). Interestingly, liraglutide treatment seemed to correct BAD pathophysiology by reducing the serum concentrations of 7α -hydroxy-4-cholesten-3-one (C4), a surrogate marker of bile acid synthesis (12), and by improving bile acid retention (assessed by the 75 selenium-homotaurocholic acid [SeHCAT] test) (12). By contrast, colestevlam treatment resulted in an excessive fecal excretion of bile acids, increased plasma concentration of C4, and reduced concentration of the enterocyte-derived fibroblast growth factor 19 (FGF19), an inhibitor of bile acid synthesis secreted from enterocytes in response to intestinal bile acid absorption and activation of the nuclear farnesoid X receptor (12). Thus, albeit both colestevlam and liraglutide may alleviate BAD symptoms, their modes of action seem to be quite different.

In this article, we report predefined secondary endpoints from our head-to-head comparison of liraglutide and colestevlam for the treatment of BAD describing changes in serum and fecal concentrations of individual bile acids as well as the fecal microbiome composition to better understand the 2 drugs' modes of action.

METHODS

Participants and procedures

We included data from 52 individuals with idiopathic or postcholecystectomy, SeHCAT-verified moderate-to-severe BAD, as previously described (12,17). Participants were recruited among newly diagnosed patients from Copenhagen University Hospital—Herlev and Gentofte, Hellerup, Denmark, and from Copenhagen University Hospital—North Zealand, Hillerød,

Denmark. The study was approved by the Danish Medicines Agency and the Regional Committee on Health Research Ethics of the Capital Region of Denmark and registered at the Danish Data Protection Agency (17). The trial was performed in accordance with good clinical practice and adhered to the newest version of the Helsinki Declaration. The eligibility criteria have been reported in detail previously (17); key criteria included a SeHCAT test result $\leq 10\%$, no diabetes, and no other gastrointestinal diseases. In short, procedures involved a run-in period of at least 1 week with no BAD treatment, after which the participants were randomly assigned to 6 weeks of double-blinded, double-dummy liraglutide treatment (once-daily subcutaneous [sc.] injection uptitrated by weekly increments of 0.6 mg from 0.6 to 1.8 mg over 3 weeks) or orally administered colesevelam treatment (1,875 mg twice daily). A daily stool diary was filled out. At baseline (just before treatment initiation), after 3 weeks of treatment, and at trial completion after 6 weeks of treatment, stool and fasting blood samples were collected.

Serum and fecal bile acid analyses

Fractionated serum bile acids were measured with liquid chromatography-tandem mass spectrometry as described previously (18). Quantification of individual bile acids was based on working calibrators produced from the Metabolomics Bile Acid Mixtures (Cambridge Isotope Labs, Tewksbury, MA). Fecal bile acids were extracted from fecal matter and measured as wet weight with ultra-high-performance liquid chromatography quadrupole time-of-flight tandem mass spectrometry, as described previously (12).

Gut microbiota composition

The gut microbial composition was evaluated using 16S rRNA gene amplicon sequencing with GridION x5 (Oxford Nanopore Technologies, Oxford, United Kingdom). Bacterial genomic DNA was extracted from 148 fecal samples using the Bead-Beat Micro AX Gravity Kit (A&A Biotechnology, Gdynia, Poland) following the manufacturer's instructions. Polymerase chain reaction amplification of the 16S rRNA gene, barcoding, library preparation, sequencing, and raw data treatment, including filtering, trimming, and taxonomy assignment, were performed as previously described (19). A total of 102 samples were included in the analysis, whereas the remaining 46 samples were excluded because of low DNA quality and/or low read counts. The resulting abundance table was analyzed using the Qiime2 (v2018.11) bioinformatic platform (20). Alpha diversity (observed species and Shannon indices) and beta diversity (principal coordinate analysis based on Bray-Curtis and Jaccard distance matrices) were calculated with rarefaction to 5,000 reads per sample. Permutational multivariate analysis of variance was used to test differences among the 3 groups based on Bray-Curtis and Jaccard distance matrices. The differences in taxa relative abundance were analyzed with Analysis of Composition of Microbiomes (19) and Linear discriminant analysis Effect Size (21).

Statistical analysis

In accordance with the study protocol (17), changes in serum and fecal bile acid concentrations over time were analyzed using a constrained linear mixed model (22,23) including time (categorical) and the constrained time $\times$ treatment-interaction as fixed effects and with an unstructured covariance pattern to account for repeated measurements on each study participant. All

outcomes were log-transformed before analysis because of having severely skewed distributions. Hence, results are reported as %-wise differences in geometric means with 95% confidence intervals (CI). *P* values were adjusted for multiple comparisons using the method of Benjamini and Hochberg (24), which controls the false discovery rate. An adjusted *P* value < 0.05 was considered statistically significant. All analyses were performed with R (25) version 4.2.2 and the R package LMMstar (26).

RESULTS

Study participants

Fifty-two patients with moderate-to-severe SeHCAT-verified BAD were included, of which 32 were women and 18 were men, as previously described (12). The participants were randomized 1:1 to colesevelam or liraglutide treatment; one woman dropped out of the liraglutide group, and one man dropped out of the colesevelam group (both because of gastrointestinal side effects) (12). The participants in the liraglutide group and the colesevelam group had a mean (SD) age of 51.7 (13.3) years and 48.7 (12.8) years, a mean body mass index of 31.5 (5.6) kg/m² and 28.3 (4.1) kg/m², and a mean SeHCAT of 3.6 (3.4)% and 4.4 (3.2)%, respectively (12).

Total, primary, and secondary bile acids in serum

Serum concentrations of primary, secondary, and total bile acids seemed to be unaffected by liraglutide while being slightly to moderately decreased after colesevelam treatment (Figure 1a,c,e and Table 1). After 6 weeks, secondary serum bile acids were reduced by -56% (CI -70% to -36%) with colesevelam compared with liraglutide, whereas primary and total bile acids showed nonsignificant estimated treatment differences. Correspondingly, the ratio between primary and secondary bile acids in serum doubled in response to colesevelam while remaining unchanged after liraglutide (Figure 2a and Table 1).

Total, primary, and secondary bile acids in feces

As previously reported (12), the fecal concentrations of bile acids were not affected by liraglutide treatment, but more than doubled after colesevelam treatment (Figure 1b,d,f and Table 1).

Total unconjugated and conjugated bile acids in serum

Colesevelam reduced the concentrations of conjugated and taurine-conjugated bile acids but no significant estimated treatment differences were detected compared with liraglutide after 6 weeks (Figure 3c,e and Table 1). Liraglutide treatment increased the serum concentration of unconjugated bile acids by 55% (CI 11%–116%) after 3 weeks (Figure 3a) and decreased the serum concentration of conjugated bile acids insignificantly (Figure 3c,e,g and Table 1). Combined, this resulted in a shift in the ratio between serum unconjugated and conjugated bile acids from consisting of mostly conjugated bile acids at baseline to primarily unconjugated bile acids after 6 weeks of liraglutide treatment (Figure 2b and Table 1). This increase in the serum concentration of unconjugated bile acids after 3 weeks of liraglutide treatment was evident in all the most abundant bile acid species measured, i.e., cholic acid, chenodeoxycholic acid, and deoxycholic acid (Figures 4c and 5c and 6c).

Total unconjugated and conjugated bile acids in feces

After 6 weeks, fecal concentrations of unconjugated, total conjugated, and glycine-conjugated bile acids were highly increased

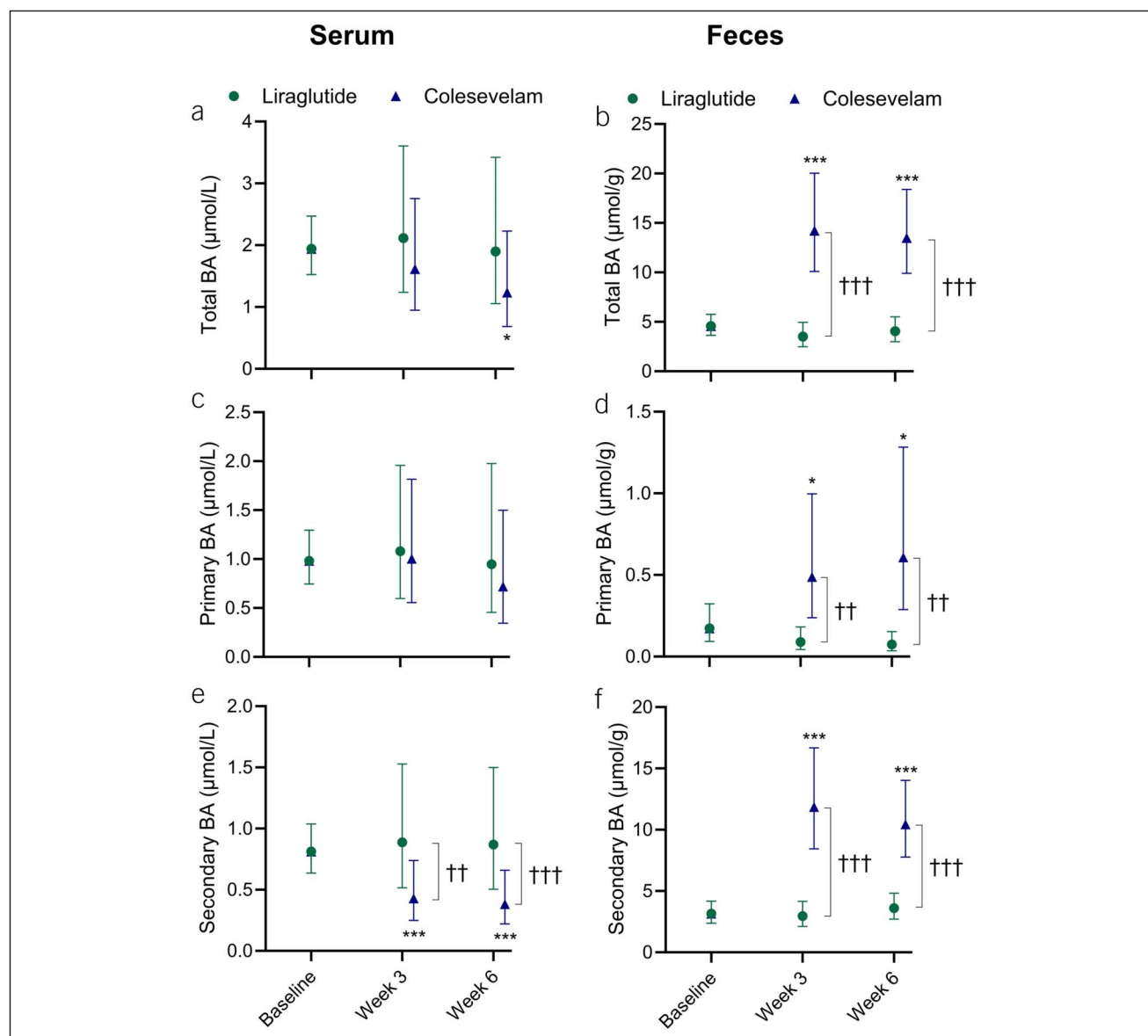


Figure 1. Total, primary, and secondary bile acids. Effect of 3 and 6 weeks of treatment with liraglutide (green circles) or colesevelam (blue triangles) on the levels of total bile acids (a, b), primary bile acids (c, d), and secondary bile acids (e, f) in serum (a, c, e) and in feces (b, d, f). Data are presented as log-transformed geometric means with 95% confidence intervals; *P* values are adjusted by the Benjamini and Hochberg procedure; asterisks represent statistical difference between data points at week 3 or 6 compared with baseline, daggers represent statistical significance in the estimated treatment difference; */†*P* < 0.05; **/††*P* < 0.01; ***/†††*P* < 0.001.

by colesevelam treatment but unchanged by liraglutide treatment (Figure 3b,d,h and Table 1).

Primary bile acids: cholic acid and chenodeoxycholic acid in serum

Colesevelam increased the concentrations of taurine-conjugated cholic acid and glycine-conjugated cholic acid compared with liraglutide after 6 weeks (Figure 4e,g and Table 1). The concentrations of unconjugated and glycine-conjugated chenodeoxycholic acid were reduced by colesevelam compared with liraglutide after 6 weeks (Figure 5c,e and Table 1). The only change observed with liraglutide

treatment was the above-mentioned increase in the serum concentration of unconjugated chenodeoxycholic acid (Figure 5c and Table 1).

Primary bile acids: cholic acid and chenodeoxycholic acid in feces

Fecal concentrations of total, unconjugated, and glycine-conjugated cholic acid were increased by colesevelam compared with liraglutide (Figure 4b,d,h and Table 1). A similar pattern was observed for fecal concentrations of chenodeoxycholic acid (Figure 5b,d,f and Table 1).

Table 1. Serum and fecal concentrations of bile acids and estimated treatment differences after 6 weeks of treatment with liraglutide or colesevelam

	Serum bile acids				Fecal bile acids			
	Baseline (mean), $\mu\text{mol/L}^a$	Change with liraglutide treatment, %	Change with colesevelam treatment, %	Estimated treatment difference	Baseline (mean), $\mu\text{mol/g}$	Change with liraglutide treatment, %	Change with colesevelam treatment, %	Estimated treatment difference
Total bile acids	1.94 (1.53 to 2.47)	−2.26 (−31.0 to 38.4)	−36.4 (−55.1 to −9.88)	−34.9 (−58.9 to 3.11) $P = 0.121$	4.6 (3.6 to 5.8)	−11.3 (−36.2 to 23.2)	195 (111 to 311)	232 (119 to 405) $P < 0.001$
Primary bile acids	0.98 (0.74 to 1.30)	−3.54 (−39.0 to 52.5)	−26.8 (−53.7 to 15.7)	−24.1 (−59.7 to 43.0) $P = 0.516$	0.173 (0.093 to 0.323)	−57.4 (−82.2 to 2.02)	253 (45.2 to 747)	723 (194 to 2,208) $P < 0.001$
Secondary bile acids	0.81 (0.64 to 1.04)	7.04 (−20.7 to 44.5)	−52.9 (−65.1 to −36.5)	−56.0 (−69.8 to −36.0) $P < 0.001$	3.15 (2.38 to 4.18)	14.5 (−18.5 to 60.8)	232 (135 to 366)	189 (95.0 to 329) $P < 0.001$
Unconjugated bile acids	0.68 (0.53 to 0.87)	48.1 (1.54 to 116)	−33.3 (−54.3 to −2.66)	−55.0 (−71.9 to −28.0) $P = 0.003$	4.42 (3.49 to 5.60)	−10.8 (−36.4 to 25.2)	175 (96.0 to 288)	209 (101 to 375) $P < 0.001$
Conjugated bile acids	1.08 (0.83 to 1.41)	−36.6 (−58.1 to −4.16)	−40.2 (−60.5 to −9.61)	−5.67 (−46.8 to 67.2) $P = 0.904$	0.083 (0.064 to 0.107)	−4.56 (−42.9 to 59.5)	654 (348 to 1,168)	687 (302 to 1,441) $P < 0.001$
Taurine-conj. bile acids	0.09 (0.06 to 0.13)	−36.7 (−58.5 to −3.34)	−49.5 (−66.9 to −23.0)	−20.4 (−53.3 to 35.9) $P = 0.526$	ND	ND	ND	ND
Glycine-conj. bile acids	0.97 (0.75 to 1.24)	−25.8 (−51.2 to 12.8)	−38.0 (−59.0 to −6.29)	−16.5 (−52.5 to 47.2) $P = 0.646$	0.089 (0.048 to 0.165)	0.4 (−39.6 to −66.7)	629 (336 to 1,119)	626 (272 to 1,319) $P = 0.028$
Cholic acid								
Total	0.24 (0.18 to 0.32)	14.5 (−27.2 to 80.2)	42.9 (−9.20 to 125)	24.7 (−32.4 to 130) $P = 0.602$	0.085 (0.043 to 0.166)	−55.3 (−82.2 to 12.2)	148 (−1.93 to 530)	456 (88.1 to 1,543) $P = 0.006$
Unconj.	0.05 (0.03 to 0.08)	126 (26.2 to 306)	14.5 (−36.8 to 105)	−49.4 (−75.7 to 5.2) $P = 0.121$	0.074 (0.036 to 0.151)	−58.8 (−84.4 to 9.01)	142 (−9.39 to 546)	487 (87.2 to 1740) $P = 0.008$
T-conj.	0.01 (0.01 to 0.02)	−18.8 (−44.1 to 17.8)	82.4 (26.1 to 164)	125 (39.8 to 261) $P = 0.003$	ND	ND	ND	ND
G-conj.	0.14 (0.10 to 0.18)	−26.5 (−50.3 to 8.72)	64.7 (−10.7 to 142)	124 (33.2 to 278) $P = 0.008$	0.003 (0.002 to 0.004)	−5.48 (−41.7 to 53.4)	339 (170 to 621)	367 (155 to 755) $P < 0.001$
Chenodeoxycholic acid								
Total	0.73 (0.55 to 0.97)	−13.9 (−46.7 to 39.2)	−46.4 (−66.8 to −13.4)	−37.8 (−38.2 to 21.8) $P = 0.245$	0.074 (0.042 to 0.131)	−56.1 (−81.0 to 1.13)	301 (72.8 to 830)	815 (235 to 2,408) $P < 0.001$
Unconj.	0.11 (0.07 to 0.17)	75.8 (−1.70 to 214)	−31.3 (−61.3 to 22.0)	−60.9 (−81.0 to −19.7) $P = 0.028$	0.062 (0.033 to 0.115)	−62.2 (−84.6 to −7.09)	256 (43.9 to 786)	845 (219 to 2,696) $P < 0.001$
T-conj.	ND	ND	ND	ND	ND	ND	ND	ND
G-conj.	0.05 (0.03 to 0.07)	−30.8 (−55.8 to 8.52)	−68.3 (−79.6 to −50.5)	−54.2 (−75.3 to −15.1) $P = 0.032$	0.004 (0.003 to 0.005)	−17.2 (−53.9 to 48.7)	1,220 (632 to 2,281)	1,493 (626 to 3,395) $P < 0.001$
Deoxycholic acid								

Table 1. (continued)

	Serum bile acids				Fecal bile acids			
	Baseline (mean), $\mu\text{mol/L}^a$	Change with liraglutide treatment, %	Change with colessevelam treatment, %	Estimated treatment difference	Baseline (mean), $\mu\text{mol/g}$	Change with liraglutide treatment, %	Change with colessevelam treatment, %	Estimated treatment difference
Total	0.57 (0.41 to 0.78)	5.50 (−32.8 to 65.7)	−60.4 (−74.8 to −37.8)	−62.5 (−79.7 to −30.7) $P = 0.006$	2.13 (1.54 to 2.95)	13.5 (−20.5 to 62.1)	218 (123 to 354)	180 (79.9 to 337) $P < 0.001$
Unconj.	0.30 (0.22 to 0.41)	61.4 (13.4 to 130)	−44.8 (−61.1 to −21.7)	−65.8 (−78.0 to −46.8) $P < 0.001$	2.02 (1.45 to 2.81)	14.7 (−20.5 to 65.7)	198 (106 to 331)	160 (64.9 to 309) $P < 0.001$
T-conj.	0.02 (0.01 to 0.03)	−48.1 (−69.0 to −13.1)	−82.4 (−89.9 to −69.7)	−66.3 (−82.4 to −35.2) $P = 0.004$	ND	ND	ND	ND
G-conj.	0.19 (0.13 to 0.26)	7.28 (−28.2 to 60.3)	−74.3 (−82.8 to −61.7)	−76.1 (−85.6 to −60.4) $P < 0.001$	0.062 (0.045 to 0.084)	5.84 (−39.0 to 83.9)	562 (278 to 1,047)	523 (190 to 1,240) $P < 0.001$
Lithocholic acid								
Total	0.04 (0.03 to 0.05)	−2.34 (−26.9 to 30.6)	−4.57 (−28.6 to 27.6)	−2.28 (−33.0 to 42.5) $P = 0.925$	0.638 (0.415 to 0.979)	62.1 (0.92 to 160)	179 (73.3 to 348)	71.9 (0.50 to 197) $P = 0.102$
Unconj.	0.02 (0.02 to 0.03)	23.5 (−6.63 to 63.4)	−1.83 (−26.0 to 30.2)	−20.5 (−45.5 to 15.7) $P = 0.322$	0.638 (0.415 to 0.978)	62.1 (0.93 to 160)	178 (72.6 to 346)	71.3 (−0.9 to 196) $P = 0.103$
T-conj	ND	ND	ND	ND	ND	ND	ND	ND
G-conj.	0.01 (0.01 to 0.02)	−14.0 (−46.1 to 37.2)	−1.72 (−38.0 to 55.9)	14.4 (−34.0 to 98.2) $P = 0.736$	ND	ND	ND	ND
Ursodeoxycholic acid								
Total	0.09 (0.06 to 0.13)	−6.17 (−35.3 to 36.1)	−45.4 (−62.4 to −20.9)	−41.9 (−63.8 to −6.64) $P = 0.055$	0.014 (0.006 to 0.030)	−48.7 (−79.2 to 26.2)	70.9 (−30.9 to 322)	233 (1.64 to 992) $P = 0.093$
Unconj.	0.04 (0.02 to 0.05)	11.6 (−28.0 to 73.0)	−22.1 (−49.7 to 20.8)	−30.2 (−59.1 to 19.0) $P = 0.269$	0.014 (0.006 to 0.030)	−48.7 (−79.2 to 26.2)	70.9 (−30.9 to 322)	233 (1.64 to 992) $P = 0.093$
T-conj.	ND	ND	ND	ND	ND	ND	ND	ND
G-conj.	0.05 (0.03 to 0.07)	−30.8 (−55.8 to 8.52)	−68.3 (−79.6 to −50.5)	−54.2 (−75.3 to −15.1) $P = 0.032$	ND	ND	ND	ND
Ratio Primary:secondary bile acids	1.22 (0.99 to 1.51)	−19.2 (−43.2 to 15.0)	72.8 (21.4 to 146)	114 (33.4 to 243) $P = 0.006$	NA	NA	NA	NA
Ratio Unconjugated: conjugated bile acids	0.60 (0.47 to 0.75)	148 (55.0 to 297)	4.44 (−34.8 to 67.2)	−57.9 (−77.1 to −22.7) $P = 0.016$	NA	NA	NA	NA

Data are log-transformed geometric means with 95% confidence intervals in brackets. P values are adjusted by the Benjamini and Hochberg procedure; P values below 0.05 are considered statistically significant and highlighted in bold font.

G-conj, glycine-conjugated; NA, not applicable; ND, not detected; T-conj., taurine-conjugated.

^aUnit does not apply to ratios.

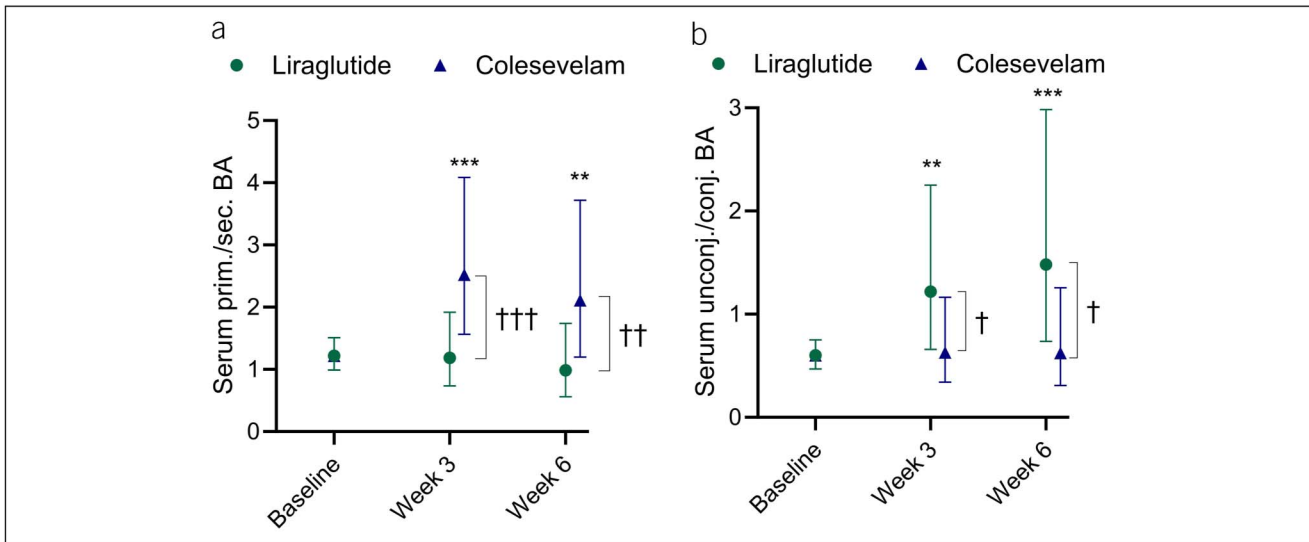


Figure 2. Ratios of serum primary/secondary and unconjugated/conjugated bile acids. Effect of 3 and 6 weeks of treatment with liraglutide (green circles) or colesevelam (blue triangles) on the ratios between primary and secondary bile acids (a) and unconjugated and conjugated bile acids (b) in serum. Data are presented as log-transformed geometric means with 95% confidence intervals; *P* values are adjusted by the Benjamini and Hochberg procedure; asterisks represent statistical difference between data points at week 3 or 6 compared with baseline, daggers represent statistical significance in the estimated treatment difference; †*P* < 0.05; **/††*P* < 0.01; ***/†††*P* < 0.001.

Secondary bile acids: deoxycholic acid, lithocholic acid, and ursodeoxycholic acid in serum

Colesevelam decreased the serum concentrations of total, unconjugated, and taurine- and glycine-conjugated deoxycholic acid, whereas liraglutide increased the concentration of unconjugated deoxycholic acid and decreased taurine-conjugated deoxycholic acid (Figure 6a,c,e,g and Table 1). Colesevelam treatment reduced total and glycine-conjugated serum ursodeoxycholic acid (see Supplementary Fig. S2A, S2E, <http://links.lww.com/CTG/B205> and Table 1), whereas liraglutide treatment did not affect serum concentrations of lithocholic acid or ursodeoxycholic acid (see Supplementary Fig. S1, Fig. S2, <http://links.lww.com/CTG/B205> and Table 1).

Secondary bile acids: deoxycholic acid, lithocholic acid, and ursodeoxycholic acid in feces

Colesevelam increased the fecal concentrations of total, unconjugated, and glycine-conjugated deoxycholic acid, whereas these seemed unaffected by liraglutide (Figure 6b,d,h and Table 1). After 6 weeks, total, unconjugated, and glycine-conjugated deoxycholic acid was increased by colesevelam treatment compared with liraglutide treatment. Colesevelam treatment increased total and unconjugated fecal lithocholic acid (see Supplementary Figure S1B, S1D, <http://links.lww.com/CTG/B205> and Table 1), but neither lithocholic acid nor ursodeoxycholic acid concentrations were affected by liraglutide treatment (see Supplementary Figure S1, Figure S2, <http://links.lww.com/CTG/B205> and Table 1).

Fecal microbiome

The analysis of the 16S rRNA gene amplicon sequencing data indicated no effect on fecal microbiome composition (permutational multivariate analysis of variance based on Bray-Curtis distance matrix; a total of 102 samples were included in the analysis, whereas the remaining 46 samples were excluded

because of low DNA quality and/or low read counts) (Figure 7a). Furthermore, analysis at baseline (*n* = 30), 3 weeks (*n* = 20 in the liraglutide group and *n* = 17 in the colesevelam group), or 6 weeks (*n* = 20 in the liraglutide group and *n* = 15 in the colesevelam group) after treatment separately showed no effect. Both the Chao1 and Shannon diversity index (Figure 7b) did not differ with treatment. Finally, because of relatively high variation and low statistical power, no particular bacterial taxa could be identified as responsible for that shift (Analysis of Composition of Microbiomes and Linear discriminant analysis Effect Size).

DISCUSSION

In this article, we report that colesevelam results in a significant increase in the fecal concentration of most bile acid species accompanied by a reduction in the serum concentrations of several bile acid species. By contrast, liraglutide did not affect fecal bile acid concentrations but induced a small but statistically significant increase in the serum concentrations of unconjugated bile acids, alluding to distinct modes of action of the 2 drugs.

In accordance with previous findings (11–13), colesevelam increased the fecal excretion of all the measured bile acid species, apart from ursodeoxycholic acid, which was found in very low concentrations. This, combined with the decreased serum concentrations of several bile acid species confirms that intraluminal sequestration of bile acids by colesevelam prevents their reabsorption. Specifically, the decrease in serum secondary bile acids and conjugated bile acids observed with colesevelam treatment indicates that primary and conjugated bile acids bound by the sequestrant in the proximal small intestine become less available for bacterial dehydroxylation and deconjugation in ileum and colon. Furthermore, previous *in vitro* binding studies have shown that bile acid sequestrants have a higher affinity for dihydroxylated bile acids (i.e., chenodeoxycholic acid and deoxycholic acid) compared with trihydroxylated bile acids (i.e., cholic acid) (27–29), which *in vivo* translates into an increase in trihydroxylated bile acids at the expense of a decrease in

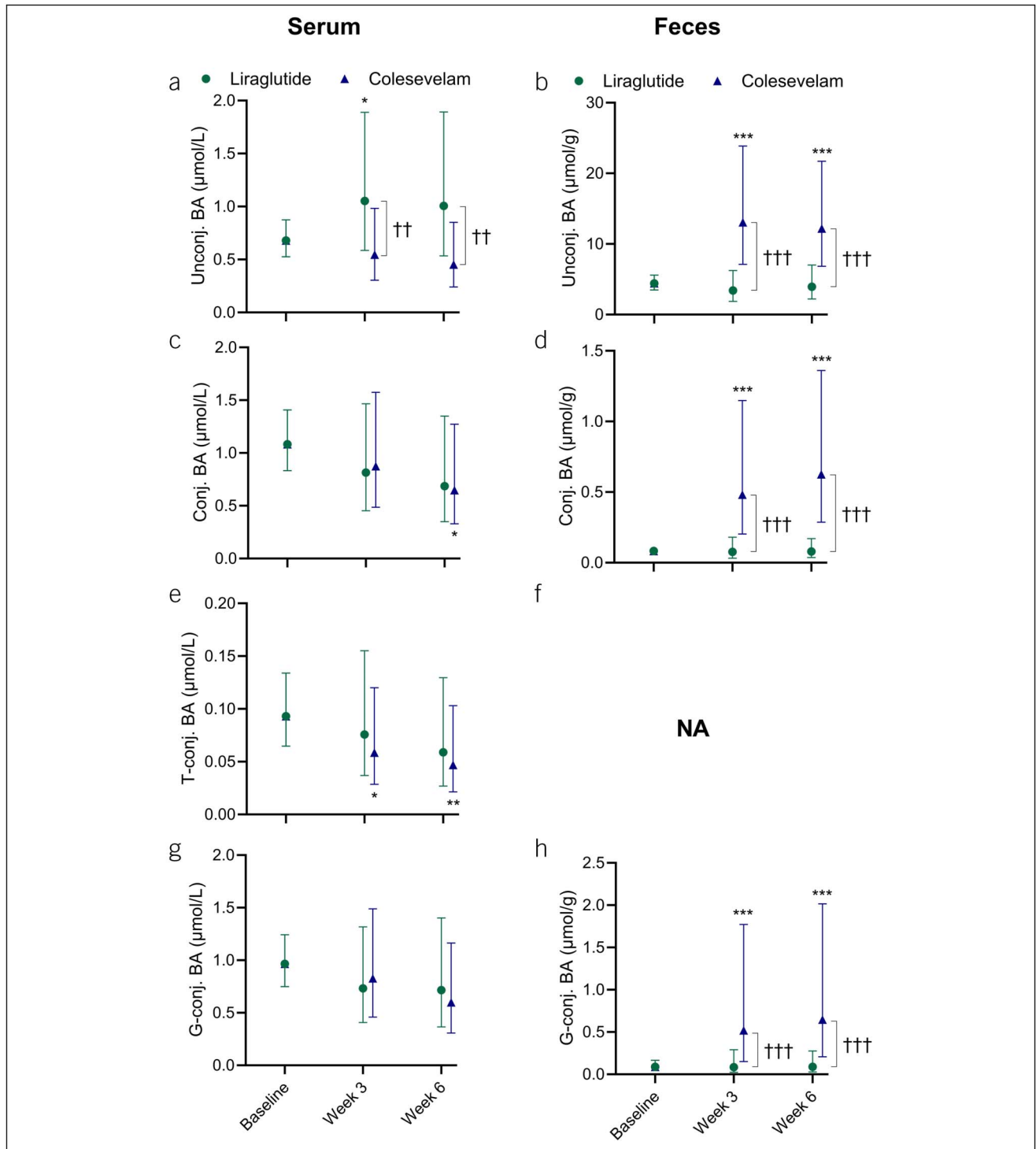


Figure 3. Unconjugated and conjugated bile acids. Effect of 3 and 6 weeks of treatment with liraglutide (green circles) or colessevelam (blue triangles) on the levels of unconjugated bile acids (a, b), conjugated bile acids (c, d), taurine-conjugated (T-conj.) bile acids (e, f), and glycine-conjugated (G-conj.) bile acids (g, h) in serum (a, c, e, g) and in feces (b, d, f, h). Data are presented as log-transformed geometric means with 95% confidence intervals; *P* values are adjusted by the Benjamini and Hochberg procedure; asterisks represent statistical difference between data points at week 3 or 6 compared with baseline, daggers represent statistical significance in the estimated treatment difference; **P* < 0.05; **/*†††P* < 0.01; ***/*†††P* < 0.001. BA, bile acid; NA, not available.

dehydroxylated bile acids (30). Accordingly, we observe decreased serum chenodeoxycholic acid and deoxycholic acid, and increased serum cholic acid with colessevelam treatment suggesting that the dehydroxylated but to a lesser extent trihydroxylated bile acids

become unavailable for absorption. In line with the overall colessevelam-induced reduction of bile acid absorption, we and others have previously reported that colessevelam treatment reduces the plasma concentration of FGF19 (secreted from enterocytes in

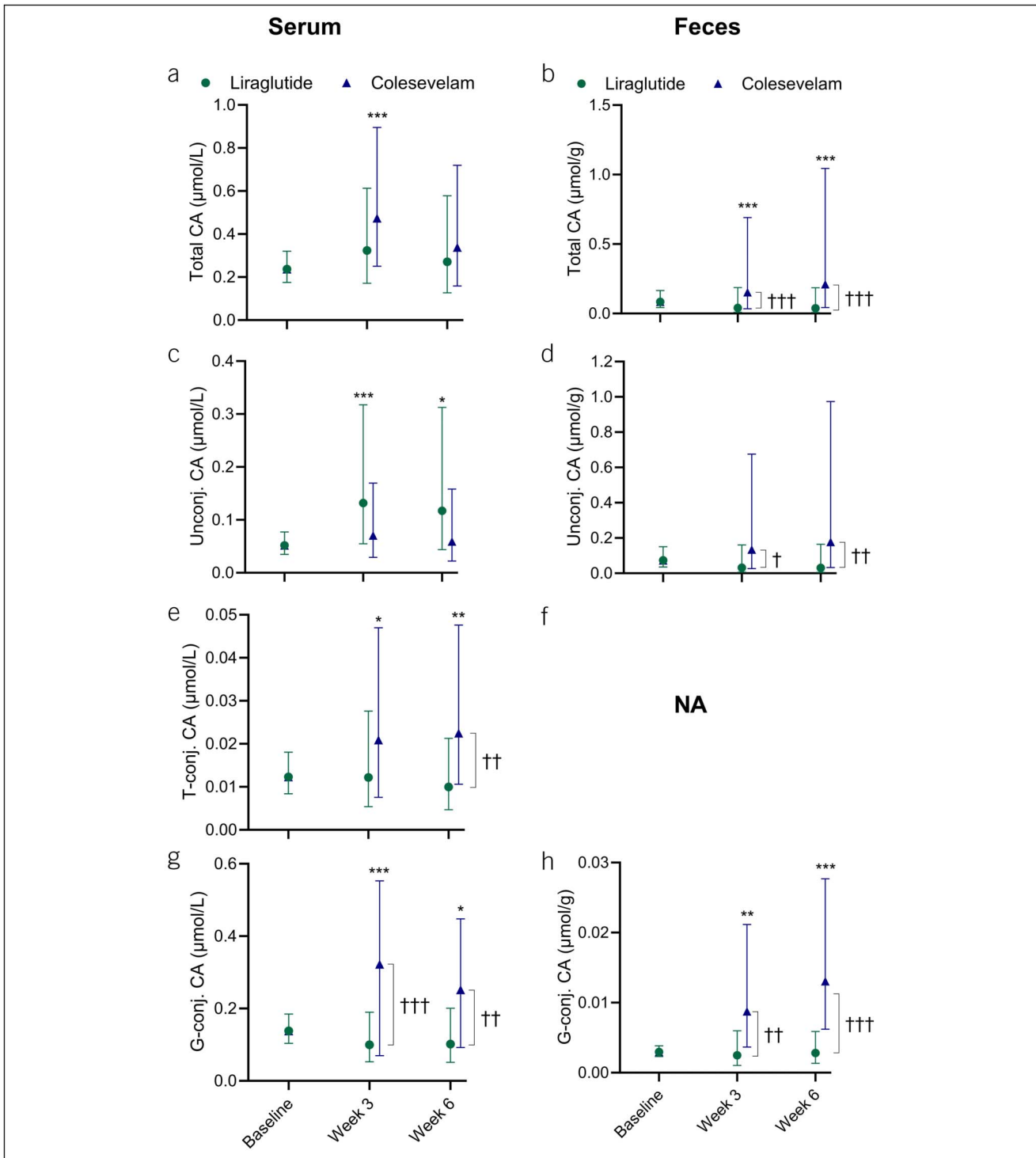


Figure 4. Cholic acid. Effect of 3 and 6 weeks of treatment with liraglutide (green circles) or colesevelam (blue triangles) on the levels of cholic acid (a, b), unconjugated cholic acid (c, d), taurine-conjugated (T-conj.) cholic acid (e, f), and glycine-conjugated (G-conj.) cholic acid (g, h) in serum (a, c, e, g) and in feces (b, d, f, h). Data are presented as log-transformed geometric means with 95% confidence intervals; *P* values are adjusted by the Benjamini and Hochberg procedure; asterisks represent statistical difference between data points at week 3 or 6 compared with baseline, daggers represent statistical significance in the estimated treatment difference; */†*P* < 0.05; **/††*P* < 0.01; ***/†††*P* < 0.001. CA, cholic acid; NA, not available.

response to intestinal bile acid absorption and activation of the nuclear farnesoid X receptor) (12,13). Correspondingly, the reduced concentration of circulating bile acids and FGF19—both potent inhibitors of hepatic bile acid synthesis—explains the colesevelam-

induced increase in plasma C4 concentrations, a biomarker for hepatic bile acid synthesis (11,12).

Treatment for 6 weeks with liraglutide did not alter the bile acid concentrations in feces. In serum, however, the concentrations of

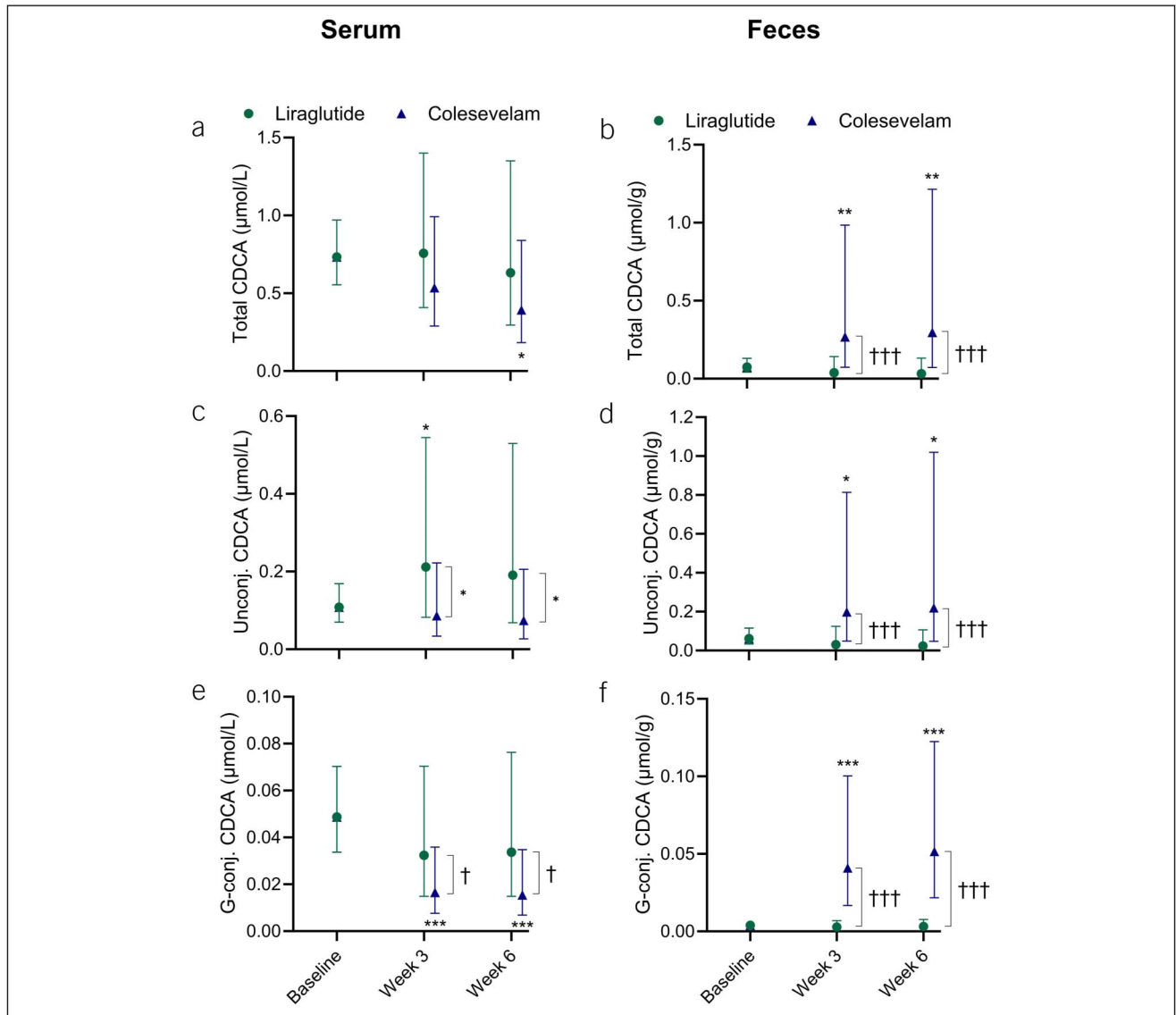


Figure 5. Chenodeoxycholic acid. Effect of 3 and 6 weeks of treatment with liraglutide (green circles) or colesevelam (blue triangles) on the levels of chenodeoxycholic acid (**a, b**), unconjugated chenodeoxycholic acid (**c, d**), and glycine-conjugated (G-conj.) chenodeoxycholic acid (**e, f**) in serum (**a, c, e**) and in feces (**b, d, f**). Taurin-conjugated chenodeoxycholic acid was below detection range in both serum and feces. Data are presented as log-transformed geometric means with 95% confidence intervals; *P* values are adjusted by the Benjamini and Hochberg procedure; asterisks represent statistical difference between data points at week 3 or 6 compared with baseline, daggers represent statistical significance in the estimated treatment difference; */ $\dagger P < 0.05$; **/ $\dagger\dagger P < 0.01$; ***/ $\dagger\dagger\dagger P < 0.001$. CDCA, chenodeoxycholic acid; NA, not available.

unconjugated cholic acid, chenodeoxycholic acid, and deoxycholic acid were significantly elevated. Deconjugation and dehydroxylation of bile acids in the small intestine are facilitated by bacterial enzymes and result in an increased acid dissociation constant (pKa) making the bile acids less water-soluble, which in turn increases their passive absorption because of reduced ionization. This is in line with the previously reported liraglutide-induced improvement in bile acid retention assessed by SeHCAT test (12). Studies show that liraglutide increases the gastrointestinal transit time (31,32). Hence, we speculate that the increased intestinal transit time induced by liraglutide treatment allows for a longer time for the bacterial enzymes to deconjugate the bile acids leading to a larger pool of unconjugated bile acids in the lumen and a longer time for passive absorption of

these bile acids. The exact mechanisms by which glucagon-like peptide 1 receptor agonism inhibits gut motility remains unclear but may involve direct actions on the enteric nervous system or intestinal smooth muscle cells (33). Furthermore, these increases in plasma concentrations of bile acids may contribute to the reduced plasma concentration of C4 observed with liraglutide treatment (12). Taken together, liraglutide treatment seems to exert a corrective impact on BAD pathophysiology by increasing absorption and decreasing synthesis of bile acids.

In our study, the gut microbiome was unaffected by the treatment with both colesevelam and liraglutide as assessed by alpha and beta diversity analyses as well as taxonomic analysis. This is in contrast with a recent study by Kumar et al (15) which

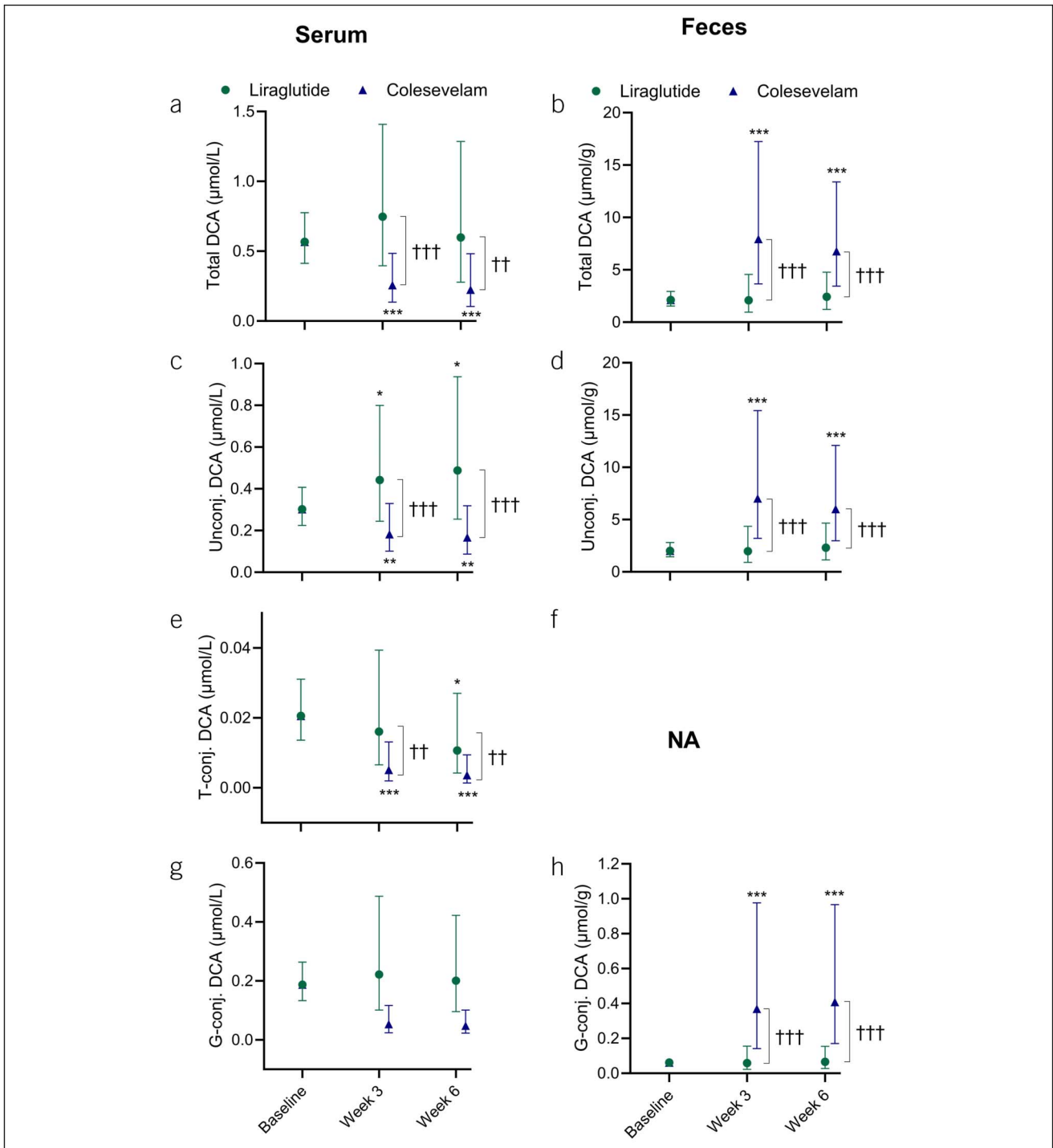


Figure 6. Deoxycholic acid. Effect of 3 and 6 weeks of treatment with liraglutide (green circles) or colesevelam (blue triangles) on the levels of deoxycholic acid (**a, b**), unconjugated deoxycholic acid (**c, d**), taurine-conjugated (T-conj.) deoxycholic acid (**e, f**), and glycine-conjugated (G-conj.) deoxycholic acid (**g, h**) in serum (**a, c, e, g**) and in feces (**b, d, f, h**). Data are presented as log-transformed geometric means with 95% confidence intervals; *P* values are adjusted by the Benjamini and Hochberg procedure; asterisks represent statistical difference between data points at week 3 or 6 compared with baseline, daggers represent statistical significance in the estimated treatment difference; * $\dagger P < 0.05$; ** $\dagger\dagger P < 0.01$; *** $\dagger\dagger\dagger P < 0.001$. DCA, deoxycholic acid; NA, not available.

reported a taxonomic shift after 8 weeks of colesevelam treatment in patients with idiopathic BAD. Interestingly, they observed no effect after 4 weeks of treatment, indicating that it takes longer time than 4 weeks for the effect of treatment to manifest, which

might explain the discrepancy between our and their findings. Moreover, our study enrolled less than 20% of the number of patients participating in the study by Kumar et al (15). Further research is needed to understand the effects of microbial

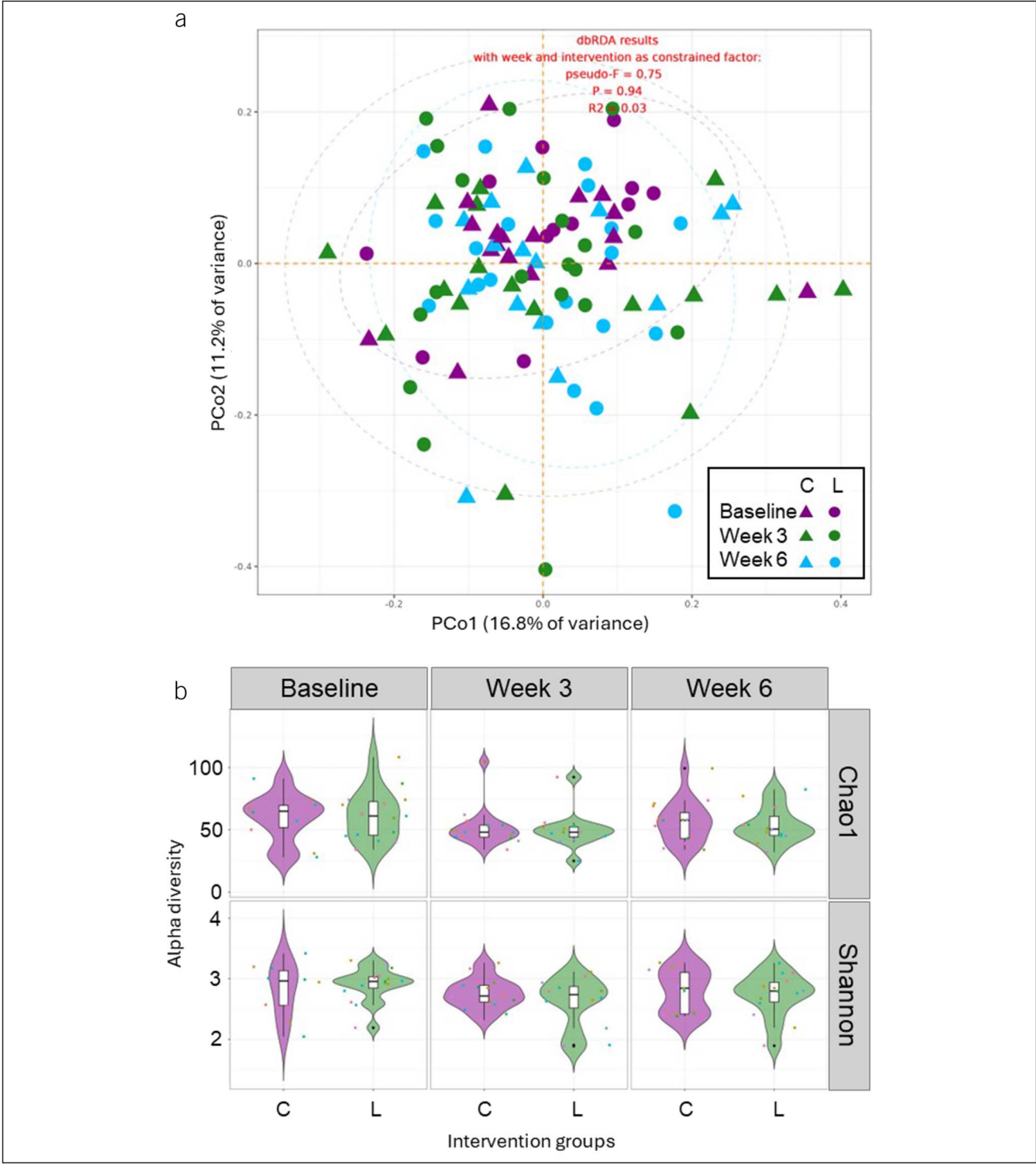


Figure 7. Microbial beta and alpha diversity analysis. Principal coordinate analysis plot based on the Bray-Curtis dissimilarity metric (a), indicating no significant differences in relation to time or treatment (liraglutide and colesevelam). Differences were tested using a distance-based redundancy analysis, with results presented in the plot (highlighted in red font). Bacterial species richness and alpha diversity (b) based on Chao1 and Shannon indices in patients treated with liraglutide or colesevelam. (c) colesevelam; dbRDA, distance-based redundancy analysis; L, liraglutide; PCo, principal coordinate.

composition on bile acid metabolism and *vice versa* in terms of response to different types of treatment.

This study was not powered for the secondary endpoints reported here; as such, any lack of significant differences in the present data needs to be interpreted cautiously. Furthermore, serum and fecal concentrations of bile acids are affected by a plethora of different stimuli (such as diet, pH, and metabolism), which were not evaluated in this study. Moreover, the fecal

weights were not assessed, and consequently, the data do not allow for evaluation of total bile acids secreted. In addition, the study only included patients suffering from primary or post-cholecystectomy BAD; thus, whether the observations from this study apply to patients with secondary BAD remains unresolved.

In conclusion, colesevelam and liraglutide both alleviate BAD symptoms (12) but seem to have distinct modes of action. Colesevelam increased fecal bile acid excretion leading to reduced bile acid absorption and serum FGF19 concentrations increasing bile acid synthesis, whereas liraglutide treatment elevated circulating concentrations of unconjugated bile acids, known to reduce bile acid synthesis (as previously observed during liraglutide treatment of BAD).

CONFLICTS OF INTEREST

Guarantors of the article: Anne-Marie Ellegaard, MSc, PhD, and Filip K. Knop, MD, PhD.

Specific author contributions: M.L.K., D.P.S., J.L.F., and F.K.K.: planned the study; M.L.K.: collected the data; A.M.E., M.L.K., D.P.S., J.L.F., S.H.H., L.K., L.O.D., D.S.N., and F.K.K.: analyzed and interpreted the data; A.M.E., L.K., and F.K.K.: drafted the manuscript; all authors revised the manuscript and approved the final draft submitted.

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Trial registration: EudraCT, 2018-003575-34.

Study Highlights

WHAT IS KNOWN

- ✓ Six weeks of treatment with colesevelam and liraglutide, respectively, improves bile acid diarrhea symptoms.
- ✓ Colesevelam and liraglutide's modes of action differ: Colesevelam binds bile acids in the gut lumen, whereas liraglutide decelerates small intestinal transit time.

WHAT IS NEW HERE

- ✓ Evaluation of serum and fecal bile acid concentrations reveals that colesevelam enhances excretion of all bile acid species and reduces absorption of secondary bile acids.
- ✓ Liraglutide treatment increases absorption of unconjugated bile acids.
- ✓ Fecal microbiome analysis shows no difference in microbiome composition with either treatment.

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